

RouteForge BDD Studio

v1.0 Enterprise

Data-Agnostic Behavioral Requirement Engineering
For Business Analysts, Architects, & QA Engineers

ISO 20022 XML

SWIFT MT103

ACH FlatFile

JSON

YAML

Jira Export

□ Why Enterprise BAs Love RouteForge:

1. Complete Data Freedom: XML, SWIFT, FlatFiles, JSON directly in Gherkin.
2. 1-Click Permutation Simulation: Real-time <placeholder> variable substitution.
3. Automatic Jira Story & Sub-Task Generator (Wiki, Markdown, JSON).
4. Local Directory Sync: Scan local .feature folders & subfolder trees.
5. Zero Dependencies: Runs pure client-side with 100% data privacy.

Why Conventional BDD Tools Fail

In Banking & Enterprise FinTech Systems

❑ The Pain Points of Legacy BDD Tools:

- Rigid JSON Lock-in: Enterprise banking runs on ISO 20022 XML & SWIFT MT.
- Hard to Model FlatFiles: NACHA ACH & ISO 8583 settlement flat lines fail.
- Manual Jira Copy-Paste: BAs waste hours converting Gherkin to Jira Stories.
- Clunky Mocking: Hard to verify payload variable substitution dynamically.

❑ The RouteForge BDD Studio Solution:

- Format-Agnostic DocStrings (""xml, ""swift, ""flatfile, ""json).
- 1-Click Sample Presets: ISO 20022 pacs.008 XML & SWIFT MT103 templates.
- Automated Jira Generator: 1-click Wiki Markup & Sub-Task export.
- Dynamic Interactive Simulator: 1-click data table substitution.

Multi-Format Data Payload Editor

ISO 20022 XML • SWIFT MT103 • ACH FlatFiles • JSON • YAML

The screenshot displays the RouteForge BDD Studio interface. On the left, the 'PROJECT SIDEBAR' shows a project named 'SWIFT Payment Flow' with a tree structure containing 'Features', 'Payloads', 'Schemas', and 'Scenarios'. The 'payment_processing.feature' file is selected. The main editor area shows the Gherkin feature file content:

```
1 Feature: Cross-Border Payment Processing
2
3 Scenario: Process SWIFT MT103 and ISO 20022 XML Payment
4   Given a valid ISO 20022 XML instruction is received
5   When the payment is validated using Payload Editor
6   And the equivalent SWIFT MT103 block is generated
7   Then the payment state should be set to "PROCESSED"
8   And a JSON response confirms the transaction.
```

Below the Gherkin editor is the 'PAYLOAD EDITOR' section, which is divided into three tabs: 'ISO 20022 XML', 'SWIFT MT103', and 'JSON'. The 'ISO 20022 XML' tab is active, showing the XML payload for the given step. The 'SWIFT MT103' tab shows the corresponding SWIFT MT103 block, and the 'JSON' tab shows the JSON response.

ISO 20022 XML:

```
1 <Document xmlns="urn:iso:std:iso:20022:tech:..."
2   <FIToFICstmrCdtTrf>
3     <GrpHdr>
4       <MsgId>RF987654321</MsgId>
5       <CreDtTm>2024-05-15T10:30:00Z</CreDtTm>
6     </GrpHdr>
7     <CdtTrfTxInf>
8       <PmtId>
9         <InstrId>INST1001</InstrId>
10        <EndToEndId>E2E2001</EndToEndId>
11      </PmtId>
12      <IntrBkSttlmAmt Ccy="USD">50000.00</IntrBkSttlmAmt>
13    </CdtTrfTxInf>
14  </FIToFICstmrCdtTrf>
15 </Document>
```

SWIFT MT103:

```
1 :20:INST1001
2 :32A:240515USD50000,00
3 :50K:/US123456789
4 JANE DOE
5 456 OAK ST
6 NEW YORK
7 :59:/DE987654321
8 JOHN SMITH
9 :71A:SHA
```

JSON:

```
1 {
2   "transactionId": "RF987654321"
3   "status": "PROCESSED",
4   "amount": {
5     "value": 50000.00,
6     "currency": "USD"
7   },
8   "timestamps": {
9     "processed": "2024-05-15T10:30:00Z"
10  }
11 }
```

The bottom status bar indicates 'Project: Payment Flow' and 'Status: Validated'.

⚡ Key Capability:

Business Analysts specify real XML tags, SWIFT blocks, or ACH lines directly inside Gherkin step containers with 1-click sample template insertion.

1-Click Permutation Simulation

Real-Time <placeholder> Variable Substitution & Pass/Fail Status

Dashboard

Scenarios

History

Reports

BDD SIMULATION RUNNER - Scenario: Submit SEPA Payment

PASSRun SimRestartSettings

1 GivenPASSED

User "Alice" is authenticated

2 WhenPASSED

User initiates a SEPA Payment for 1500 EUR

3 ThenPASSED

The transaction payload is generated and substituted

AndPENDING

The request is sent to the Payment Gateway

AndPENDING

The payment status is confirmed

PAYLOAD GENERATION & SUBSTITUTION (Step 3)

The transaction payload is generated and substituted.

SUBSTITUTED XML PAYLOAD (ISO 20022 pain.001)PASS

SUBSTITUTED SWIFT PAYLOAD (MT103)PASS

SIMULATION LOG

Generating XML/SWIFT payloads... Substitutions applied: Transaction ID [TXN-00123-ABC], Amount [1500.00 EUR]. (0.04s) [PASSED]

SIMULATION CONTEXT

Variables

TransactionID: TXN-00123-ABC

Environment: Production

Schema: ISO 20022 v09

Execution Metadata

Execution ISO: 137055

Execution Metadata

TransactionID: TXN-00123-ABC

Environment: 1500

Execution time: 2024-08-28

⚡ **Step-by-Step Test Engine:**
Executes Gherkin scenario steps in sequence. Replaces <placeholder> variables in XML and JSON payloads with data table row values in real time.

Architected by Pratyush Ranjan Mishra

<https://geekpratyush.github.io/BDDModeler>

Jira User Story & Sub-Task Generator

1-Click Agile Ticket Creation (Wiki Markup • Markdown • REST JSON)

Dashboard | Home | Jira

Project AI Dev Studio

Projects

Boards

Issues

Reports

Settings

Dashboard

Jira Story & Sub-Task Generator

Title: Enhance User Profile Page | Issue Type: Story | Priority: High | Status: To Do

Description | Acceptance Criteria | Sub-Tasks

Wiki Markup | Markdown

****Goal:**** Users must be able to update their profile picture to enhance all profile pic you : and user a that manailtes more monlw information.

****Jira Wiki:****

```
{panel:title=User Profile Enhancement}
*As a user, I want...*
```

[Copy Wiki Markup] [Copy Markdown]

Acceptance Criteria

- ☒ User can upload new JPG/PNG
- ☒ File size limit (2MB) handled
- ☐ Profile pic updates immediately... [Add Criteria]

Sub-Task Checklist

- ☒ Create backend API endpoint (Story point: 3)
- ☒ Implement frontend file upload component (Story point: 5)
- ☐ Add validation rules (Story point: 2)
- ☐ Write unit tests (Story point: 3)
- ☐ UI/UX design review (Story point: 1)

[Add Sub-Task]

Generate Jira Content | Save Changes | Copy All | Create Issue

⚡ Zero Manual Copy-Pasting:

Automatically converts BDD scenarios into Jira Acceptance Criteria with code blocks and auto-generated Sub-Tasks. 1-click Copy to Clipboard & Download.

Folder Groups & Custom Domains

Organize Complex Project Subfolders & Industry Templates

▣ Subfolder Group Management

- Native File System directory scan
- Recursive mapping of local .feature folders
- Dedicated '+ Group' button in sidebar
- Safe '× Group' folder deletion
- Workspace Tabs Bar for open files

▣ Custom Domain Engine

- Built-in Domains: Banking, Retail, Health
- '+ New Domain' custom industry creator
- E.g. Insurance, Freight, Telecom
- Persistent localStorage storage
- Instant domain workspace switching

▣ Pre-Built Banking Templates Included:

1. Intraday Clearing Line Utilization & Limit Queueing
2. Excess Approval & Hierarchy Dual Signoff Escalation
3. Daily Overdraft & Limit/Sublimit Earmarking Releases
4. ISO 20022 MX pacs.008 XML to SWIFT MT103 Message Transformation
5. Sanctions Screening & OFAC Quarantine Engine

Try RouteForge BDD Studio Live!

100% Free & Open-Source Zero-Dependency Tool

☐ Experience the Live Application:

- ☐ <https://geekpratyush.github.io/BDDModeler/>
- ☐ GitHub: <https://github.com/geekpratyush/BDDModeler>
- ☐ LinkedIn: <https://www.linkedin.com/in/leadtherightway/>

☐☐ About the Author & Architect:

Pratyush Ranjan Mishra

Enterprise Systems Architect & Requirement Engineering Specialist

I design zero-friction tools for Business Analysts and Engineering teams to model complex payment flows, ISO 20022 payloads, and BDD specifications.

☐ **Connect on LinkedIn to share feedback or collaborate on BDD tools!**